

Database Connection Architecture

This is a .NET 8.0 application using a **multi-tenant database connection architecture** with Oracle.

Technology Stack

- **ORM:** Entity Framework Core 8.0.8
- **Database:** Oracle (using `Oracle.ManagedDataAccess.Core` and `Oracle.EntityFrameworkCore`)

Connection Flow

1. **Factory Pattern** - The application uses `IDigiLiSDbContextFactory` to create database contexts dynamically per tenant:
 - `Cloud4Log.Http/DigiLiS/Data/DbContexts/DigiLiSDbContextFactory.cs:15`
2. **Connection String Retrieval** - Connection strings are fetched from **Google Secret Manager** at runtime:

```
var connectionString = await googleSecretManagerService.GetLatestSecretVersionAsync
```

3. **Oracle Connection Creation** - A new Oracle connection is created with the retrieved connection string:

```
var oracleConnection = new OracleConnection(connectionString);  
return optionsBuilder.UseOracle(oracleConnection).Options;
```

4. **Schema-aware DbContext** - The `DigiLiSDbContext` is created with a dynamic schema name (format: `DIGILIS{company}{branch}`):
 - `Cloud4Log.Http/DigiLiS/Data/DbContexts/DigiLiSDbContext.cs:3`

Multi-tenant Pattern

The database identifier format is `0-{company}-{branch}` (e.g., "O-10-33"). This maps to:

- Secret name: `DIGILIS-10-33`
- Schema name: `DIGILIS1033`

Key Components

Component	Location
Factory Interface	DigiLiS/Data/DbContexts/Intefaces/IDigiLiSDbContextFactory.cs
Factory Implementation	DigiLiS/Data/DbContexts/DigiLiSDbContextFactory.cs
DbContext	DigiLiS/Data/DbContexts/DigiLiSDbContext.cs
Cache Key Factory	DigiLiS/Data/DbContexts/DigiLiSDbContextCacheKeyFactory.cs
Secret Manager Service	GoogLeCloudPlatform/SecretManager/GoogLeSecretManagerService.cs
DI Setup	Infrastructure/SetupExtensions/DigiLiSSetupExtension.cs

Usage in Services

Services inject `IDigiLiSDbContextProvider` and get a context for a specific tenant:

```
var digiLiSContext = await dbContextProvider.GetDbContext(databaseIdentifier);
```